

Adaptive Computing in Federated Tactical Clouds

Overview, Results and Follow-Up for IST-193 RTG "Edge Computing at the Tactical Edge"

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ABSTRACT

Enhanced military operations increasingly rely on advanced sensing, processing, storage, and communication, despite challenges posed by limited bandwidth and connectivity in Tactical Edge Environments (TEEs). The NATO IST-168 project developed a federated tactical cloud architecture and evaluated Kubernetes-based cloud orchestrators for TEEs. Extending on this, the IST-193 project elaborated the four fundamental services for tactical clouds to be federated and enabling adaptive workload distribution across them: Federated Resource Discovery, Federated Service Deployment, Trust Resource Management, and Federated Adaptive Orchestrator. Performance assessments indicate the feasibility of these adaptive solutions in TEEs. This paper summarizes the IST-193 and IST-168 architecture work, together with performance and cost-benefit analysis of adaptive solutions. Future research topics are presented that will focus on enhanced (large-scale) performance analysis, advanced adaptivity scenarios, robust trust and security mechanisms, and adoption of the federated and adaptive tactical cloud architectures and concepts.

1.0 INTRODUCTION

Military missions are increasingly data-driven, relying on advanced sensing, processing, storage, and communication capabilities. However, these capabilities are often operating in Tactical Edge Environments (TEEs) and interconnected over disadvantaged Tactical Networks (TNs) with limited bandwidth, intermittent connectivity, and variable latency. Federated and adaptive tactical clouds can potentially offer significant advantages in joint coalition missions operating in TEEs, both for military operations (in terms of improved situational awareness and mission effectiveness) and for IT operations (in terms of data processing efficiency and infrastructure resilience). The conventional centralized cloud infrastructures, while effective in civilian applications, are unsuitable for such disadvantaged TEEs. A comparison of the cloud computing capabilities in the civilian and in the military context are provided in Figure 1, together with an overview of the networking technologies as typically used in (the various levels of) a disadvantaged TEE.

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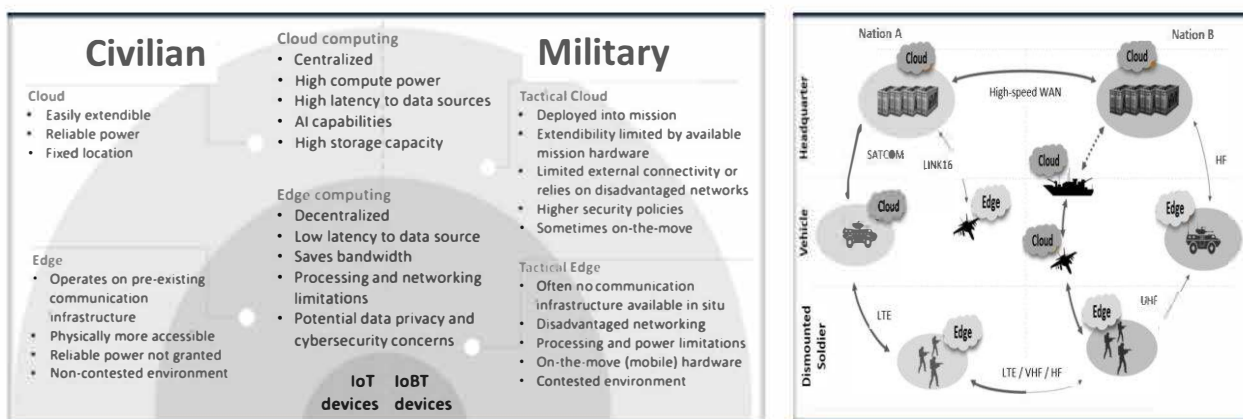


Figure 1: Cloud computing in civilian versus military context (l) and networking technologies as typically used in (the various levels of) a disadvantaged TEE (r).

The NATO IST-168 Research Task Group (IST-168 RTG) on "*Adaptive Information Processing and Distribution to Support Command and Control*" has developed a federated Tactical Cloud (TC) architecture that distributes resources and processing tasks among coalition partners, providing a resilient and adaptable solution for joint operations in TEEs. Moreover, it assessed the performance of various commercial off-the-shelf, Kubernetes-based, cloud orchestrator distributions suitable for low-bandwidth environments. Building upon the findings of the IST-168 RTG, the NATO IST-193 RTG on "*Edge Computing at the Tactical Edge*" has further developed the architecture into a set of four services enabling the TCs to be federated, whilst allowing each individual tactical cloud to retain its sovereignty. The findings of the IST-168 and IST-193 RTGs suggest that a federated and adaptive cloud approach can be used in disadvantaged TNs (despite their networking constraints) and offer major data processing benefits.

This paper summarizes the results of the IST-168 and IST-193 RTGs as previously published in [1] - [7], together with topics for further research. The high-level architecture for the federated and adaptive tactical clouds for a TEE is described in section 2.0. Subsequently, section 3.0 addresses the role of workload deployment orchestration and summarizes the associated Kubernetes and adaptivity cost-benefit performance assessment results. Finally, Section 4.0 provides the overarching conclusions and outlines the topics future work.

2.0 FEDERATED AND ADAPTIVE CLOUD ARCHITECTURE FOR A TEE

The architecture for a military TEE, as explored in the IST-168 and IST-193 RTGs, extends beyond the traditional notions of cloud computing in civilian contexts, which typically involve a centralized pool of resources managed within a secure, high-bandwidth environment. Military operations, however, require a more decentralized and dynamic approach due to the disadvantaged TN environments, for which a federated and adaptive cloud architecture is proposed as depicted in Figure 2 [6][7].

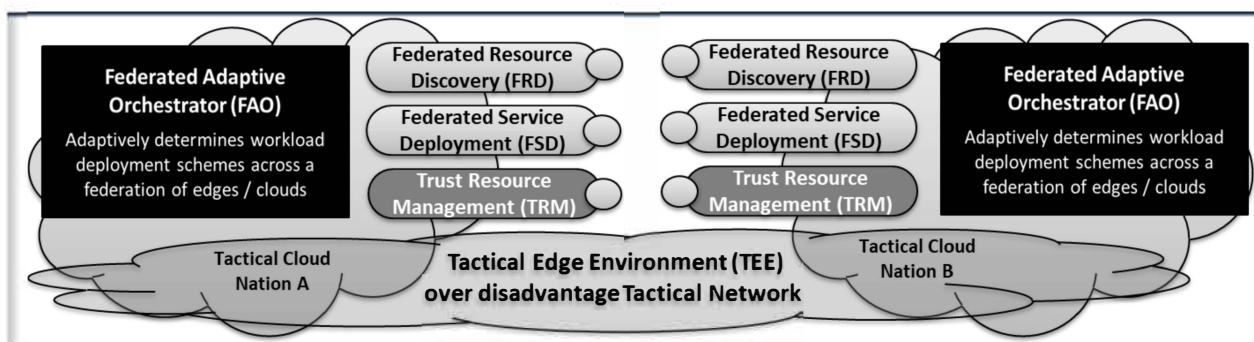


Figure 2: Federated cloud architecture with adaptive workload orchestration for disadvantaged TEEs.

As depicted in Figure 1, tactical clouds (equipped with computational power and storage capabilities) run on edge computing devices close to the soldiers on the battlefield, for instance in military vehicles in the TEE. By federating these tactical clouds, resource availability and adaptability are bolstered, with minimal or no dependence on access to a centralised cloud. Federation allows NATO allies to mutually use each other's tactical cloud infrastructures by sharing computing workloads, making the most of processing power without overloading communication. This may enable near real-time analysis capabilities at the tactical edge. e.g. for sophisticated analytics tools to extract actionable insights, including pattern recognition and predictive modelling, or to support decision-making effectively and dynamically.

Moreover, this sharing of computing workloads is done in an adaptive manner. Both the communications network and the computing workload status are continuously monitored. Computing workloads are re-distributed when conditions change, e.g. when adversary actions disable a partner's cloud or when the bandwidth of a communication link diminishes as a vehicle with a cloud instance moves out-of-range. This adaptive approach of sharing computing workloads across the federation of tactical clouds is realised while each of the mission partners preserves autonomy over its own tactical clouds: each mission partner can define and enforce its own policy rules and conditions on exposing its computing resources to mission partners. This is done by means of four architectural services, available on each of the partner's clouds as shown in Figure 2 [6][7]:

- The *Federated Resource Discovery service* that provides information on both available services (available for deployment and already deployed) and resources (e.g., CPUs, RAM, storage). It queries its own cloud and other available clouds to collect and aggregate this information. Additionally, it exposes resource and service information about its own cloud through an external API after applying any policy constraints.
- The *Federated Adaptive Orchestrator service* that is responsible for creating deployment plans for computing workloads across the federation. These plans are developed based on the information provided by the Federated Resource Discovery service. The Federated Adaptive Orchestrator does not expose an external API.
- The *Federated Service Deployment service* that is responsible for deploying computing workloads either on its own cloud or on other available clouds. It exposes an external API with policy constraints for communicating between the Federated Service Deployment services of all clouds.
- The *Trust Resource Management service* that collects per-partner-cloud security metrics to dynamically compute trust values for all partner clouds in the federation. These trust values can be used by the other services in various clouds to assist with decision-making, e.g. exposing resources and services or allowing specific partners to deploy computing workloads on its cloud.

Together, these four services enable joint operations of the federated and adaptive cloud infrastructure, ensuring autonomy and security for each mission partner. Leveraging open-source Kubernetes distributions specifically optimised for low-bandwidth environments, performance assessments confirm the feasibility of deploying these services in disadvantaged TEEs.

This federated and adaptive tactical cloud architecture has been demonstrated through various Proofs-of-Concept (PoCs) for illustrative and representative scenarios that demonstrate the practical application and potential benefits in a complex, real-world context. The scenarios involve NATO coalition operation in an urban setting, where multiple NATO partners collaborate to identify and track a suspicious vehicle, requiring intensive video processing with adaptive workload orchestration across the partner's tactical clouds in a disadvantaged TEE.

3.0 DEPLOYMENT ORCHESTRATION AND PERFORMANCE ASSESSMENT

Deployment orchestration in a federated cloud environment requires sophisticated mechanisms to manage the allocation of resources and distribution of processing workloads across multiple clouds in TEEs.

3.1 Deployment Orchestration

A significant challenge in deployment orchestration is maintaining service availability despite network disruptions in the disadvantaged TNs which are inherently unstable with frequent changes in connectivity and network performance. A robust orchestration mechanism must continuously monitor network conditions and dynamically reallocate processing workloads to ensure critical services remain operational.

Optimizing resource usage across diverse and distributed nodes is another critical challenge. Tactical environments are resource-constrained, with limited computational power, bandwidth, and energy. Efficient scheduling algorithms are essential for maximizing the use of available resources while minimizing latency and bandwidth consumption.

Therefore, deployment orchestration algorithms must be capable of adapting to the dynamic conditions of the network, reallocating workloads in response to changes in node availability and network performance. In the federated cloud environment, the proposed orchestration strategy involves a hybrid intra- and inter-cloud orchestration approach. Each tactical cloud manages and optimizes resource and service allocation within its own tactical clouds independently, referred to as “intra-cloud orchestration”. In addition, “inter-cloud orchestration” coordinates resource sharing and service deployment across multiple tactical clouds, ensuring efficient use of distributed capabilities while respecting the sovereignty of each coalition partner's resources.

3.2 Kubernetes Performance Assessment

The performance assessment of various Kubernetes distributions assessed by IST-RTG-168 highlights the distinct advantages and limitations of the KubeEdge, K8s, and K3s Kubernetes distributions in TNs [3][4][5]. The experiments were conducted by realistic testbed configurations simulating different experiment scenarios. An experiment was a combination of various network scenarios (LAN, BEST, AVG, WORST) with different parameters for bandwidth, latency, and packet loss, and either a cluster initialization or application deployment phase.

The experiments demonstrated that KubeEdge, leveraging the QUIC protocol, outperforms K8s and K3s in disadvantaged network conditions, showcasing superior resilience and stability. Unlike K8s and K3s, which depend on the TCP protocol and struggle under unstable network scenarios, KubeEdge maintains reliable performance, making it particularly suitable for TEE. K8s-based clusters exhibit high sensitivity to increased network latency, which significantly impacts their performance during cluster initialization and application deployment phases. Despite this sensitivity, these clusters remain relatively robust in low bandwidth environments, managing to operate effectively under the constraints of disadvantaged TNs. In terms of cluster size and network scenarios, KubeEdge demonstrates the capacity to manage larger clusters, supporting up to 18 nodes under optimal network conditions and up to 12 nodes under average conditions.

Overall, the studies affirm that KubeEdge's architecture and protocol choices make it a more viable solution for TNs, providing better performance and reliability compared to K8s and K3s, particularly in environments with fluctuating and often challenging network conditions.

3.3 Adaptivity Cost-Benefit Analysis

The cost benefit analysis by IST-RTG 193 evaluates the performance impact of adaptivity (i.e. re-allocating processing workloads) across the federated TCs in a simulated TEE [7]. Using a scenario involving vehicle recognition and tracking, the experiments measure service response times, bandwidth usage, and resource utilization for adaptive deployment of License Plate Recognition (LPR) services across federated clouds. The

setup includes multiple virtual machines configured to simulate real-world network conditions, with tools controlling bandwidth and connectivity constraints.

The results demonstrate that the adaptivity costs of implementing adaptive computing at the tactical edge are minimal. This includes low usage of bandwidth, CPU, and memory, indicating that adaptive tactical edge computing can be feasibly implemented without significant performance overhead. The analysis shows that inter-cloud interactions for resource discovery and service deployment have a negligible impact on bandwidth, and the CPU and memory utilization of the key services (Federated Resource Discovery, Federated Adaptive Orchestrator, and Federated Service Deployment) are very low.

4.0 CONCLUSIONS AND FUTURE WORK

The synthesis of the previous IST-168 and IST-193 RTG results [1] - [7], underscores the significant potential of federated and adaptive cloud architectures to transform data sharing and processing in disadvantaged tactical networks. By addressing adaptive deployment orchestration challenges, these architectures can significantly enhance the effectiveness and resilience of coalition military missions. Overall, the PoCs and experiments (on both Kubernetes and adaptivity cost-benefit) performance analysis validate the feasibility and efficiency of adaptive orchestration of workloads in federated tactical clouds.

Additional research aims to refine these strategies and ensure their successful implementation, ultimately improving the capability of coalition forces to conduct joint operations in complex and dynamic environments. As such, future work as follow-up on the IST-168 and IST-193 RTGs work, should include:

- *Enhanced performance analysis* for large scale deployment of federated and tactical clouds, considering both (1) the performance for large scale federation of tactical clouds (e.g. based on the Anglova scenario [8]), (2) the assessment of the effectiveness of various workload deployment orchestration algorithms to dynamically allocate resources across federations of tactical clouds based on real-time network conditions and resource availability [9], and (3) developing a real-life deployment scenario for further identifying challenges and applicability to emerging application platforms in the TEE, e.g. focusing on processing sensor fusion applications in the TEE.
- *Advanced adaptivity scenarios*, considering both (1) the possibilities, benefits, challenges and drawbacks over the combined adaptive tactical clouds and the adaptive tactical connectivity network [10], and (2) the inclusion of additional features in the workload deployment orchestration algorithms, specifically trust, cost / benefit performance impact and pre-emption.
- *Robust trust and security mechanisms* that can protect against the unique threats faced by tactical clouds operating in (adversary) TEEs. This includes both (1) the dynamic computation of the per-partner-cloud trust values (as part of the *Trust Resource Management service* as described in section II) to assist with decision-making on distributing and allowing adaptive computing workloads, and (2) security hardening at the application and container level and at the infrastructure deployment container orchestrator level.
- *Adoption of the federated and adaptive tactical cloud architectures and concepts*, e.g. by aligning the proposed solutions with NATO's standardization process and Federated Mission Networking (FMN) initiative, ensuring integration into NATO's evolving TN mission frameworks.

ACKNOWLEDGEMENT

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